Lecture 19: Section 5.4

Area and Properties

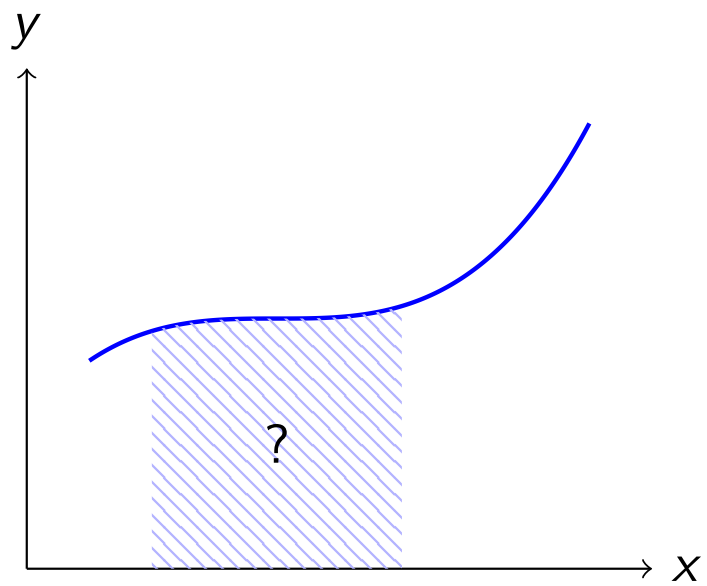
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Introduction: What is Area?

The Big Problem

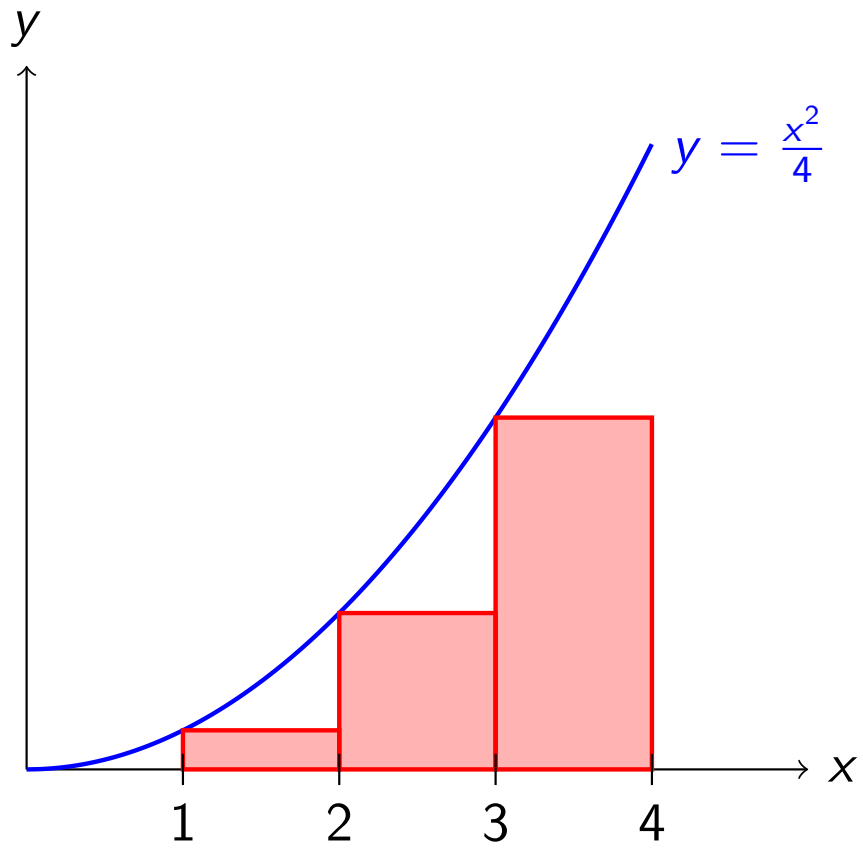
How do we find the area under a curve that isn't made of straight lines?



Our Approach

We'll approximate the area using rectangles, then take a limit as the rectangles get infinitely thin.

Area Under a Curve: The Basic Idea



Riemann Sums: Formal Definition

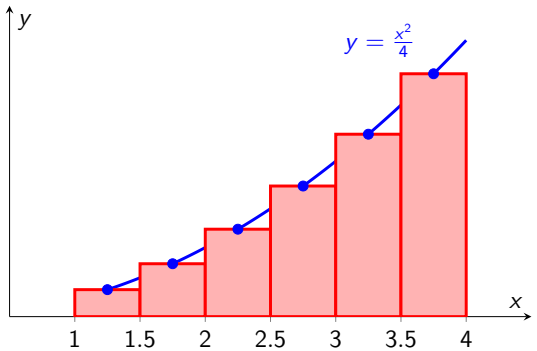
Definition 1 (Riemann Sum)

The area under $y = f(x)$ from $x = a$ to $x = b$ is approximated by:

$$A \approx \sum_{i=1}^{n} f(x_i^*)\Delta x$$

where:

- $\Delta x = \frac{b-a}{n}$ (width of each rectangle)
- x_i^* is a sample point in the ith subinterval
- n is the number of rectangles



The Exact Area as a Limit

Definition 2 (Area as a Limit)

The exact area under $y = f(x)$ from $x = a$ to $x = b$ is:

$$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$$

where $\Delta x = \frac{b-a}{n}$ and x_i^* is any point in the ith subinterval.

Why This Works

- More rectangles $\rightarrow$ better approximation
- As $n \rightarrow \infty$, $\Delta x \rightarrow 0$
- Rectangles become infinitely thin
- Sum becomes exact area

Example 3

Find area under $y = x^2$ from $x = 0$ to $x = 1$.

Example: Area Under $y = x^2$

Find area under $y = x^2$ from $x = 0$ to $x = 1$.

Solution:

1. Divide $[0, 1]$ into n equal subintervals: $\Delta x = \frac{1}{n}$
2. Use right endpoints: $x_i^* = \frac{i}{n}$
3. Height of ith rectangle: $f(x_i^*) = \left(\frac{i}{n}\right)^2$
4. Area approximation: $A_n = \sum_{i=1}^{n} \left(\frac{i}{n}\right)^2 \cdot \frac{1}{n}$
5. Simplify: $A_n = \frac{1}{n^3} \sum_{i=1}^{n} i^2 = \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6}$
6. Take limit: $A = \lim_{n\to\infty} \frac{(n+1)(2n+1)}{6n^2}$
7. Calculate: $A = \lim_{n\to\infty} \frac{2n^2+3n+1}{6n^2} = \frac{2}{6} = \frac{1}{3}$

So area $= \frac{1}{3}$.

Another Example: Area Under $y = x$

Find area under $y = x$ from $x = 0$ to $x = 2$.

Solution:

1. $\Delta x = \frac{2}{n}$
2. Right endpoints: $x_i^* = \frac{2i}{n}$
3. Height: $f(x_i^*) = \frac{2i}{n}$
4. $A_n = \sum_{i=1}^{n} \frac{2i}{n} \cdot \frac{2}{n} = \frac{4}{n^2} \sum_{i=1}^{n} i$
5. $A_n = \frac{4}{n^2} \cdot \frac{n(n+1)}{2} = \frac{2(n+1)}{n}$
6. $A = \lim_{n\to\infty} \frac{2(n+1)}{n} = 2$

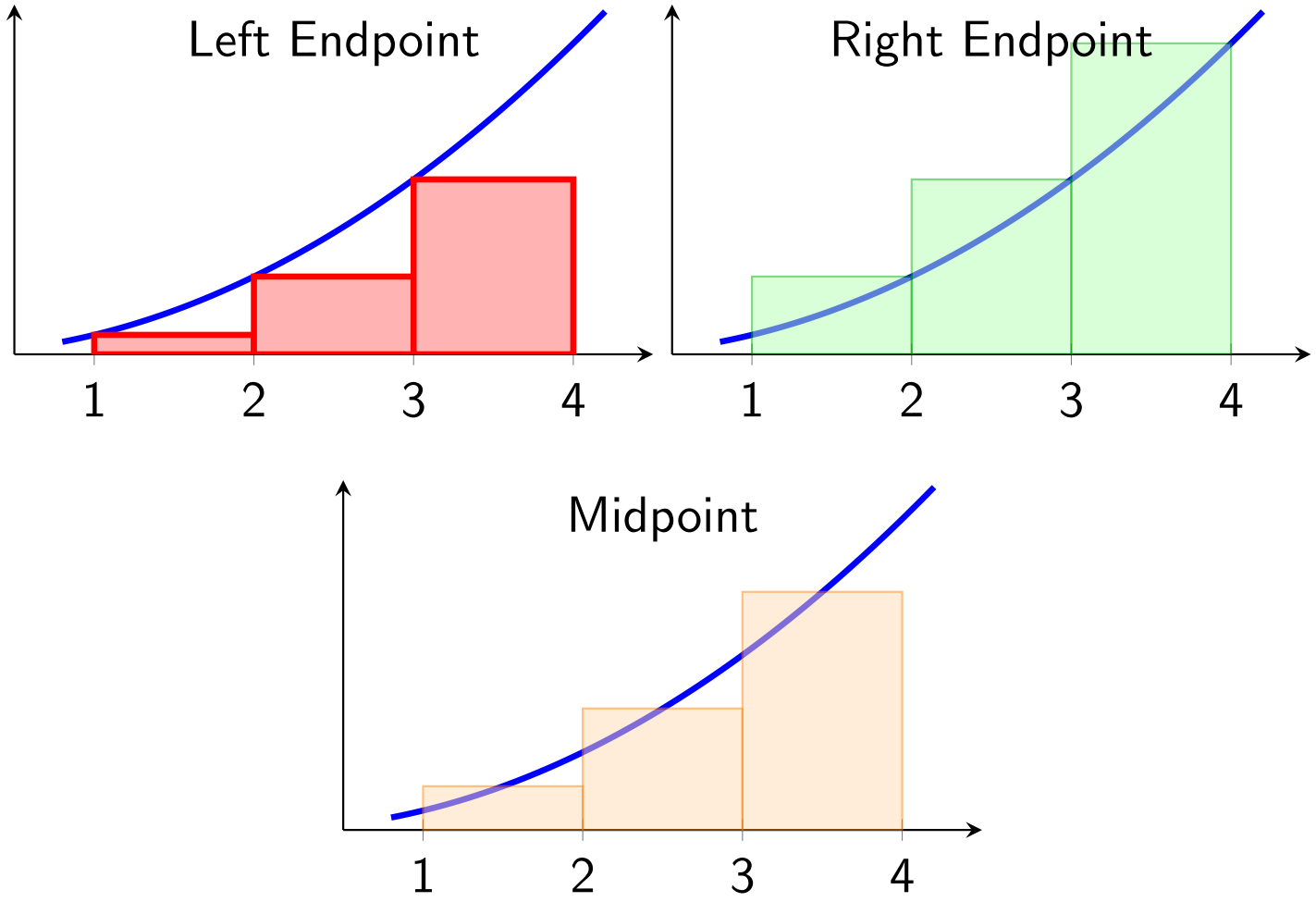
This makes sense: area of triangle with base 2 and height 2 is $\frac{1}{2} \cdot 2 \cdot 2 = 2$.

Practice: Area as a Limit (8 problems)

Set up (but don't evaluate) the limit for the area under:

1. $y = 2x$ from $x = 0$ to $x = 3$ using right endpoints
2. $y = x^2$ from $x = 1$ to $x = 2$ using left endpoints
3. $y = 3x + 1$ from $x = 0$ to $x = 4$ using right endpoints
4. $y = x^3$ from $x = 0$ to $x = 1$ using midpoints
5. $y = \sqrt{x}$ from $x = 0$ to $x = 4$ using right endpoints
6. $y = 4 - x^2$ from $x = -2$ to $x = 2$ using right endpoints
7. $y = \frac{1}{x}$ from $x = 1$ to $x = 3$ using left endpoints
8. $y = e^x$ from $x = 0$ to $x = 2$ using right endpoints

Left, Right, and Midpoint Approximations



Formulas for Approximations

Left Endpoint Approximation (L_n)

$$L_n = \sum_{i=0}^{n-1} f(x_i)\Delta x, \quad x_i = a + i\Delta x$$

Right Endpoint Approximation (R_n)

$$R_n = \sum_{i=1}^{n} f(x_i)\Delta x, \quad x_i = a + i\Delta x$$

Midpoint Approximation (M_n)

$$M_n = \sum_{i=1}^{n} f(\bar{x}_i)\Delta x, \quad \bar{x}_i = \frac{x_{i-1} + x_i}{2}$$

where $\Delta x = \frac{b-a}{n}$.

Example: Approximating Area

Approximate area under $y = x^2$ from $x = 0$ to $x = 2$ using $n = 4$:

1. $\Delta x = \frac{2-0}{4} = 0.5$
2. Subintervals: $[0, 0.5], [0.5, 1], [1, 1.5], [1.5, 2]$

Left Endpoint (L_4):

$$\begin{aligned} L_4 &= f(0) \cdot 0.5 + f(0.5) \cdot 0.5 + f(1) \cdot 0.5 + f(1.5) \cdot 0.5 \\ &= (0 + 0.25 + 1 + 2.25) \cdot 0.5 = 3.5 \cdot 0.5 = 1.75 \end{aligned}$$

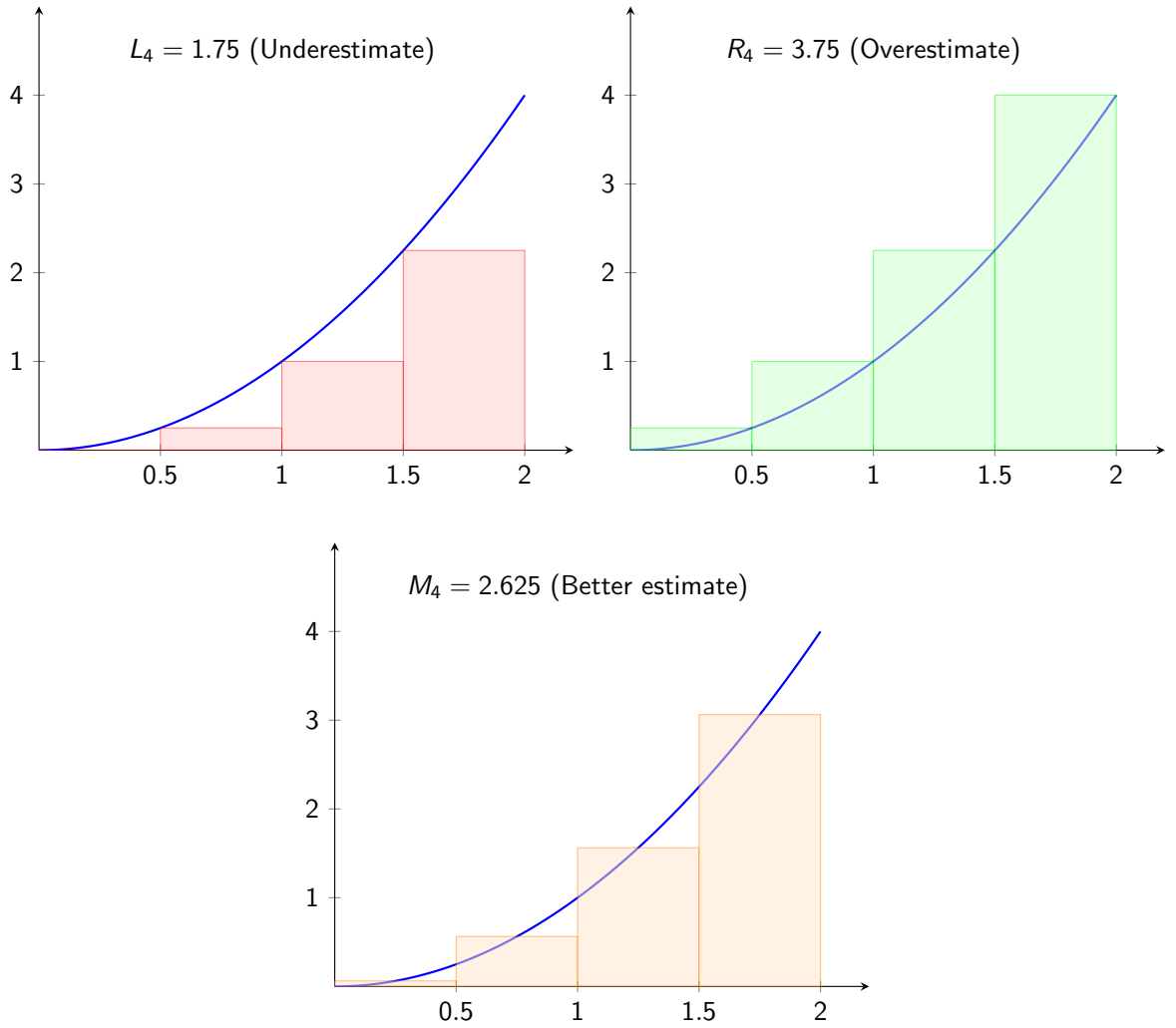
Right Endpoint (R_4):

$$\begin{aligned} R_4 &= f(0.5) \cdot 0.5 + f(1) \cdot 0.5 + f(1.5) \cdot 0.5 + f(2) \cdot 0.5 \\ &= (0.25 + 1 + 2.25 + 4) \cdot 0.5 = 7.5 \cdot 0.5 = 3.75 \end{aligned}$$

Midpoint (M_4): $\bar{x}_1 = 0.25, \bar{x}_2 = 0.75, \bar{x}_3 = 1.25, \bar{x}_4 = 1.75$

$$\begin{aligned} M_4 &= [f(0.25) + f(0.75) + f(1.25) + f(1.75)] \cdot 0.5 \\ &= (0.0625 + 0.5625 + 1.5625 + 3.0625) \cdot 0.5 = 5.25 \cdot 0.5 = 2.625 \end{aligned}$$

Which Approximation is Best?



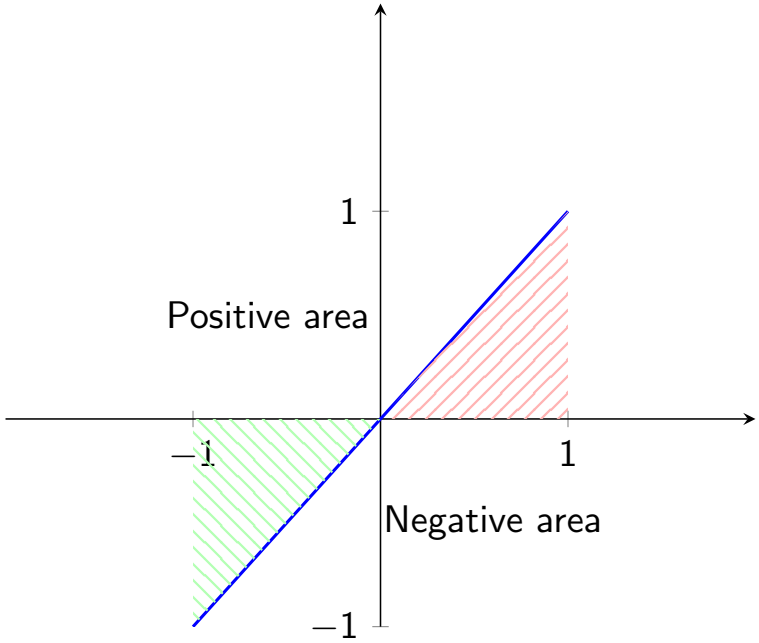
Exact area $= \frac{8}{3} \approx 2.667$ (from integration)

Practice: Numerical Approximations (10 problems)

For each, approximate area using $n = 4$ rectangles:

1. $y = 2x$ from $x = 0$ to $x = 4$, left endpoints
2. $y = x^2$ from $x = 1$ to $x = 3$, right endpoints
3. $y = 3x + 1$ from $x = 0$ to $x = 2$, midpoints
4. $y = 4 - x^2$ from $x = 0$ to $x = 2$, left endpoints
5. $y = \sqrt{x}$ from $x = 0$ to $x = 4$, right endpoints
6. $y = x^3$ from $x = 0$ to $x = 2$, left endpoints
7. $y = \frac{1}{x}$ from $x = 1$ to $x = 3$, midpoints
8. $y = e^x$ from $x = 0$ to $x = 2$, right endpoints
9. $y = \sin x$ from $x = 0$ to $x = \pi$, left endpoints
10. $y = \cos x$ from $x = 0$ to $x = \pi/2$, midpoints

What is Net Signed Area?



Definition 4 (Net Signed Area)

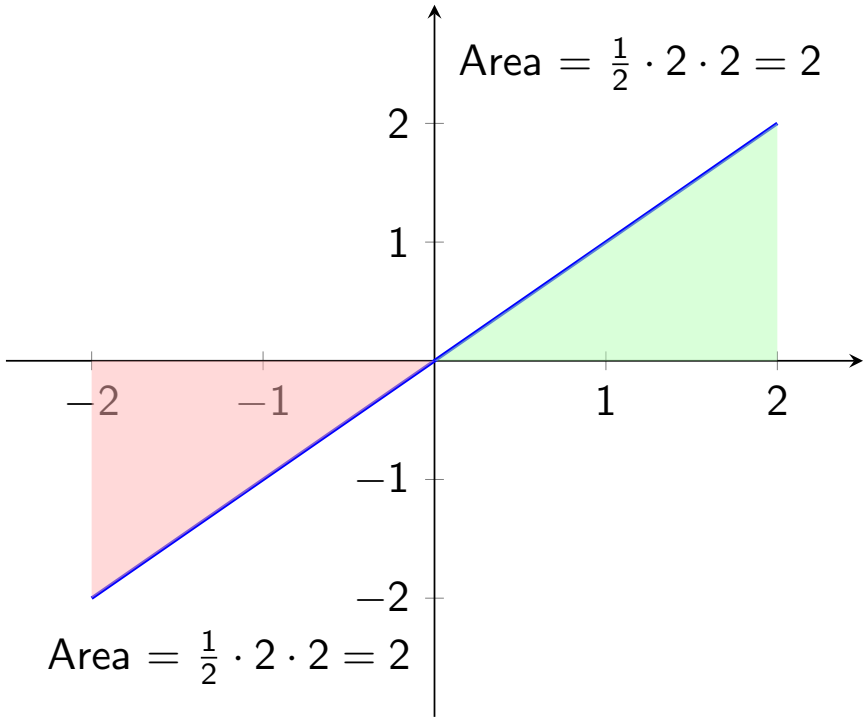
Net signed area = (Area above x-axis) - (Area below x-axis)

Key Point

When function is negative, rectangles have negative height, so their "area" is negative.

Example: Net Signed Area

Find net signed area under $y = x$ from $x = -2$ to $x = 2$.
Solution:

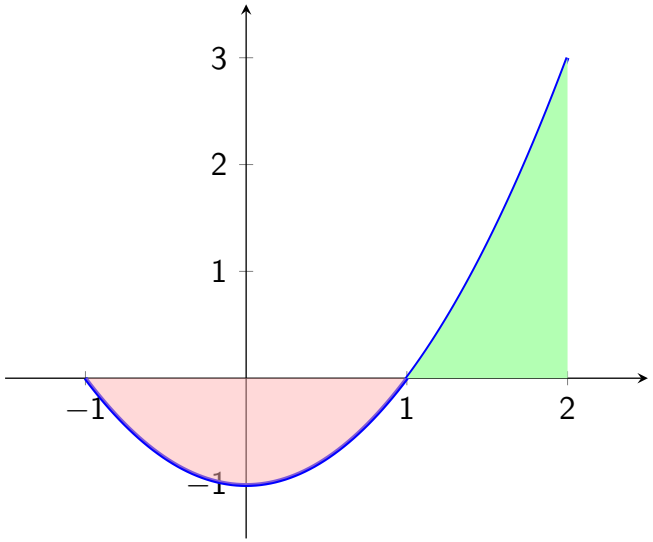


Net signed area = Area above - Area below $= 2 - 2 = 0$.
This makes sense: function is odd and symmetric about the origin.

Another Example: Net Signed Area

Find net signed area under $y = x^2 - 1$ from $x = -1$ to $x = 2$.

Solution:



Area above (from 1 to 2): $\int_1^2 (x^2 - 1)dx = \left[\frac{x^3}{3} - x\right]_1^2 = \frac{8}{3} - 2 - (\frac{1}{3} - 1) = \frac{4}{3}$

Area below (from -1 to 1):

$\int_{-1}^1 (x^2 - 1)dx = \left[\frac{x^3}{3} - x\right]_{-1}^1 = (\frac{1}{3} - 1) - (-\frac{1}{3} + 1) = -\frac{4}{3}$

Net signed area $= \frac{4}{3} + (-\frac{4}{3}) = 0$ (but total area $= \frac{8}{3}$)

Practice: Net Signed Area (8 problems)

For each, find the net signed area:

1. $y = 2x$ from $x = -1$ to $x = 2$
2. $y = x^2 - 4$ from $x = 0$ to $x = 3$
3. $y = \sin x$ from $x = 0$ to $x = 2\pi$
4. $y = x^3$ from $x = -2$ to $x = 1$
5. $y = |x|$ from $x = -2$ to $x = 2$
6. $y = \cos x$ from $x = 0$ to $x = \pi$
7. $y = 1 - x^2$ from $x = -2$ to $x = 2$
8. $y = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$ from $x = -3$ to $x = 3$

Summary: Key Concepts

1. **Sigma Notation:** Compact way to write sums
2. **Properties of Sums:** Constants factor out; sums distribute
3. **Summation Formulas:** For i, i^2, i^3
4. **Area as Limit:** $A = \lim_{n\to\infty} \sum f(x_i^*)\Delta x$
5. **Approximations:** L_n, R_n, M_n
6. **Net Signed Area:** Counts area above as positive, below as negative

Next Steps

- Practice with different functions
- Use technology to visualize approximations
- These ideas lead to definite integrals in calculus